Chapter 5: Continuity and Differentiability

School Worksheet: step-wise solutions

20 questions **51** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ — 1 mark

The value of k for which $f(x) = \begin{cases} kx + 1, & x \le 5 \\ 3x - 5, & x > 5 \end{cases}$ is continuous at $x = 5$ is:

(A) $\frac{9}{5}$ **(B)** $\frac{5}{9}$ **(C)** $\frac{11}{5}$ **(D)** 2

Formula / idea: f is continuous at $x = a$ if LHL $=$ RHL $= f(a)$.

1 Find $f(5)$ and the LHL from the first piece.
$f(5) = \lim\limits_{x \to 5^-} (kx + 1) = 5k + 1$

2 Find the RHL from the second piece.
$\lim\limits_{x \to 5^+} (3x - 5) = 15 - 5 = 10$

3 For continuity, equate them and solve. ✓ 1
$5k + 1 = 10 \Rightarrow k = \frac{9}{5}$

4 So the correct option is (A).

Answer **(A)** $\frac{9}{5}$

Watch out: Using the $x \le 5$ piece for both sides; each one-sided limit uses its own piece.

2 MCQ 1 mark

The function $f(x) = |x|$ is:

(A) continuous and differentiable everywhere

(B) **continuous everywhere but not differentiable at $x = 0$**

(C) differentiable but not continuous at $x = 0$

(D) neither continuous nor differentiable at $x = 0$

Formula / idea: Differentiable at a means LHD $=$ RHD; differentiability implies continuity, not the reverse.

1 Check continuity at $x = 0$: both one-sided limits equal $f(0)$.
$$\lim_{x \to 0^-} (-x) = 0 = \lim_{x \to 0^+} x = f(0)$$

2 Find the LHD at $x = 0$.
$$\text{LHD} = \lim_{h \to 0} \frac{|0 - h| - 0}{-h} = \lim_{h \to 0} \frac{h}{-h} = -1$$

3 Find the RHD at $x = 0$.
$$\text{RHD} = \lim_{h \to 0} \frac{|0 + h| - 0}{h} = 1$$

4 LHD $\neq$ RHD, so $|x|$ is not differentiable at 0, though continuous. ✓ 1
$$-1 \neq 1$$

5 So the correct option is (B).

Answer **(B) continuous everywhere but not differentiable at $x = 0$**

Watch out: Option (C) is impossible: a function differentiable at a point is always continuous there.

3 MCQ 1 mark

The derivative of e^{e^x} with respect to x is:

(A) $\log x$ **(B)** e^{e^x} **(C)** $e^x \cdot e^{e^x}$ **(D)** $(e^x)^2$

Formula / idea: Chain rule: $\frac{d}{dx} e^u = e^u \frac{du}{dx}$.

1 Let $y = e^{e^x}$ and apply the chain rule.
$$\frac{dy}{dx} = e^{e^x} \cdot \frac{d}{dx}(e^x)$$

2 Differentiate the inner function e^x. ✓ 1
$$\frac{dy}{dx} = e^{e^x} \cdot e^x = e^x \cdot e^{e^x}$$

3 So the correct option is (C).

Answer **(C)** $e^x \cdot e^{e^x}$

Watch out: Writing just e^{e^x} (option B) forgets to multiply by the derivative of the power e^x.

4 MCQ — 1 mark

If $y = \log(\cos e^x)$, then $\dfrac{dy}{dx}$ is:

(A) $\tan e^x$ **(B)** $-e^x \tan e^x$ **(C)** $e^x \tan e^x$ **(D)** $-\tan e^x$

Formula / idea: Chain rule, outside to inside: $\log \to \cos \to e^x$.

1 Differentiate $\log$, then $\cos$, then e^x (chain rule).

$$\frac{dy}{dx} = \frac{1}{\cos e^x} \cdot (-\sin e^x) \cdot e^x$$

2 Simplify using $\dfrac{\sin}{\cos} = \tan$. ✓ 1

$$\frac{dy}{dx} = -e^x \tan e^x$$

3 So the correct option is (B).

Answer **(B)** $-e^x \tan e^x$

Watch out: Dropping the factor e^x from the innermost function gives the wrong option (D).

5 MCQ — 1 mark

If $x = at^2$ and $y = 2at$, then $\dfrac{dy}{dx}$ is:

(A) t **(B)** $\dfrac{1}{t}$ **(C)** $2t$ **(D)** $-\dfrac{1}{t^2}$

Formula / idea: Parametric form: $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$.

1 Differentiate both w.r.t. t.

$$\frac{dx}{dt} = 2at, \qquad \frac{dy}{dt} = 2a$$

2 Divide. ✓ 1

$$\frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t}$$

3 So the correct option is (B).

Answer **(B)** $\frac{1}{t}$

Watch out: Dividing the other way round, $\frac{dx/dt}{dy/dt} = t$, gives option (A).

6 **Assertion-Reason** 1 mark

Assertion (A): The greatest integer function $f(x) = [x]$ is discontinuous at $x = 2$. **Reason (R):** $\lim_{x\to 2^-}[x] \neq \lim_{x\to 2^+}[x]$.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: A function is discontinuous at a point where LHL $\neq$ RHL.

1 Find the one-sided limits of $[x]$ at $x = 2$.
$\lim_{x\to 2^-}[x] = 1, \qquad \lim_{x\to 2^+}[x] = 2$

2 Since $1 \neq 2$, f is discontinuous at $x = 2$: **A is true**.

3 $1 \neq 2$, so **R is true**.

4 A is true precisely because LHL $\neq$ RHL, so **R explains A**. ✓ 1

5 So the correct option is (A).

Answer **(A) Both A and R are true and R is the correct explanation of A**

Watch out: Just left of 2, $[x] = 1$, not 2; students often take $[1.99] = 2$.

7 **Short answer** 2 marks

Differentiate $\sin^{-1}\left(2x\sqrt{1-x^2}\right)$ with respect to x, where $-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$.

Formula / idea: Substitute $x = \sin\theta$ to simplify, using $\sin 2\theta = 2\sin\theta\cos\theta$.

1 Let $y = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$ and put $x = \sin\theta$.
$x = \sin\theta \Rightarrow \theta = \sin^{-1}x$

2 Find the range of θ from the given range of x. ✓ ½

$$-\frac{1}{\sqrt{2}} < \sin\theta < \frac{1}{\sqrt{2}} \Rightarrow -\frac{\pi}{4} < \theta < \frac{\pi}{4}$$

3 Simplify the expression inside. ✓ ½

$$2x\sqrt{1-x^2} = 2\sin\theta\cos\theta = \sin 2\theta$$

$\cos\theta > 0$ here, so $\sqrt{1-\sin^2\theta} = \cos\theta$

4 Since $-\frac{\pi}{2} < 2\theta < \frac{\pi}{2}$, $\sin^{-1}$ cancels sin. ✓ ½

$$y = \sin^{-1}(\sin 2\theta) = 2\theta = 2\sin^{-1}x$$

5 Differentiate w.r.t. x. ✓ ½

$$\frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$$

Answer $\frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$

Watch out: Skipping the range check for 2θ; $\sin^{-1}(\sin 2\theta) = 2\theta$ only when $2\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Level 2

18 marks

8 **Short answer** 2 marks

Find k so that $f(x) = \begin{cases} \dfrac{1-\cos 4x}{8x^2}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$.

Formula / idea: Continuity at 0: $\lim\limits_{x\to 0} f(x) = f(0)$; use $\lim\limits_{\theta\to 0}\dfrac{\sin\theta}{\theta} = 1$.

1 For continuity at $x = 0$ we need:

$$\lim_{x\to 0} f(x) = f(0) = k$$

2 Use $1 - \cos 2A = 2\sin^2 A$ with $A = 2x$. ✓ ½

$$\lim_{x\to 0}\frac{1-\cos 4x}{8x^2} = \lim_{x\to 0}\frac{2\sin^2 2x}{8x^2}$$

3 Rearrange to get the standard limit $\frac{\sin 2x}{2x}$. ✓ 1

$$\lim_{x\to 0}\frac{\sin^2 2x}{4x^2} = \lim_{x\to 0}\left(\frac{\sin 2x}{2x}\right)^2 = 1^2 = 1$$

4 Equate the limit to $f(0)$. ✓ ½

$$k = 1$$

Answer $k = 1$

Watch out: Writing $1 - \cos 4x = 2\sin^2 4x$; the angle halves, so it is $2\sin^2 2x$.

9 **Short answer** 3 marks

Find $\frac{dy}{dx}$ if $y = x^{\sin x} + (\sin x)^x$.

Formula / idea: For $(\text{variable})^{\text{variable}}$, take log first; split a sum into $u + v$ before taking logs.

1	Let $u = x^{\sin x}$ and $v = (\sin x)^x$, so $y = u + v$. $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$ log of a sum cannot be split, so separate first	✓ ½
2	Take log of u. $\log u = \sin x \log x$	
3	Differentiate w.r.t. x (product rule). $\frac{1}{u}\frac{du}{dx} = \cos x \log x + \frac{\sin x}{x}$	
4	Multiply by u. $\frac{du}{dx} = x^{\sin x}\left(\cos x \log x + \frac{\sin x}{x}\right)$	✓ 1
5	Take log of v. $\log v = x \log \sin x$	
6	Differentiate w.r.t. x (product rule). $\frac{1}{v}\frac{dv}{dx} = \log \sin x + x \cdot \frac{\cos x}{\sin x}$	
7	Multiply by v. $\frac{dv}{dx} = (\sin x)^x (\log \sin x + x \cot x)$	✓ 1
8	Add the two results. $\frac{dy}{dx} = x^{\sin x}\left(\cos x \log x + \frac{\sin x}{x}\right) + (\sin x)^x (\log \sin x + x \cot x)$	✓ ½

Answer $\frac{dy}{dx} = x^{\sin x}\left(\cos x \log x + \frac{\sin x}{x}\right) + (\sin x)^x (\log \sin x + x \cot x)$

Watch out: Taking $\log y = \sin x \log x + x \log \sin x$ directly; $\log(u + v) \neq \log u + \log v$.

10 **Short answer** 2 marks

If $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$, find $\dfrac{dy}{dx}$.

Formula / idea: Parametric form: $\dfrac{dy}{dx} = \dfrac{dy/d\theta}{dx/d\theta}$.

1	Differentiate x w.r.t. θ. $\dfrac{dx}{d\theta} = a(1 + \cos\theta)$	✓ ½
2	Differentiate y w.r.t. θ. $\dfrac{dy}{d\theta} = a\sin\theta$	✓ ½
3	Divide. $\dfrac{dy}{dx} = \dfrac{a\sin\theta}{a(1+\cos\theta)}$	✓ ½
4	Use half-angle formulas and simplify. $\dfrac{dy}{dx} = \dfrac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}} = \tan\dfrac{\theta}{2}$ $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$	✓ ½

Answer $\dfrac{dy}{dx} = \tan\dfrac{\theta}{2}$

Watch out: Writing $\dfrac{d}{d\theta}(1 - \cos\theta) = -\sin\theta$; the derivative of $-\cos\theta$ is $+\sin\theta$.

11 **Short answer** 3 marks

If $y = e^{a\cos^{-1}x}$, $-1 < x < 1$, show that $(1 - x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} - a^2y = 0$.

Formula / idea: Clear the root before differentiating again; write $y_1 = \dfrac{dy}{dx}$, $y_2 = \dfrac{d^2y}{dx^2}$.

1	Differentiate y w.r.t. x (chain rule). $y_1 = e^{a\cos^{-1}x} \cdot \dfrac{-a}{\sqrt{1-x^2}} = \dfrac{-ay}{\sqrt{1-x^2}}$ $\dfrac{d}{dx}\cos^{-1}x = \dfrac{-1}{\sqrt{1-x^2}}$	✓ 1
2	Cross-multiply to remove the fraction. $\sqrt{1-x^2}\,y_1 = -ay$	✓ ½
3	Differentiate again w.r.t. x (product rule). $\sqrt{1-x^2}\,y_2 + y_1 \cdot \dfrac{-x}{\sqrt{1-x^2}} = -ay_1$	✓ 1

4	Multiply both sides by $\sqrt{1-x^2}$. $(1-x^2)y_2 - xy_1 = -a\sqrt{1-x^2}\, y_1$	
5	Put $\sqrt{1-x^2}\, y_1 = -ay$ on the right. $(1-x^2)y_2 - xy_1 = -a(-ay) = a^2y$	
6	Rearrange. Hence proved. $(1-x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} - a^2y = 0$	✓ ½

Answer $(1-x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} - a^2y = 0$ **(proved)**

Watch out: Forgetting the minus sign in the derivative of $\cos^{-1}x$, which flips the sign of a^2y.

12 **Short answer** 3 marks

If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, show that $\dfrac{dy}{dx} = \sqrt{\dfrac{1-y^2}{1-x^2}}$.

Formula / idea: Put $x = \sin A$, $y = \sin B$; then use sum-to-product formulas.

1	Put $x = \sin A$ and $y = \sin B$. $A = \sin^{-1}x, \qquad B = \sin^{-1}y$	
2	Substitute in the given equation. $\cos A + \cos B = a(\sin A - \sin B)$	✓ ½
3	Use $\cos C + \cos D$ and $\sin C - \sin D$ formulas. $2\cos\frac{A+B}{2}\cos\frac{A-B}{2} = a \cdot 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$	✓ 1
4	Cancel $2\cos\frac{A+B}{2}$ from both sides. $\cot\dfrac{A-B}{2} = a$	
5	So $A-B$ is a constant. $\sin^{-1}x - \sin^{-1}y = k \quad (k \text{ a constant})$ $\cot\frac{A-B}{2}$ equals the fixed number a	✓ ½
6	Differentiate both sides w.r.t. x. $\dfrac{1}{\sqrt{1-x^2}} - \dfrac{1}{\sqrt{1-y^2}} \cdot \dfrac{dy}{dx} = 0$ The derivative of the constant k is 0	✓ ½

7 Solve for $\frac{dy}{dx}$. Hence proved. ✓ ½

$$\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}} = \sqrt{\frac{1-y^2}{1-x^2}}$$

Answer $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$ **(proved)**

Watch out: Differentiating the original equation directly; it works, but the algebra gets long and error-prone.

13 Short answer — 2 marks

Find $\frac{dy}{dx}$ if $x^2 + 2xy + y^3 = 42$.

Formula / idea: Implicit differentiation: differentiate each term w.r.t. x, treating y as a function of x.

1 Differentiate both sides w.r.t. x. ✓ 1

$$2x + 2\left(x\frac{dy}{dx} + y\right) + 3y^2\frac{dy}{dx} = 0$$

Product rule on $2xy$; chain rule on y^3

2 Collect the $\frac{dy}{dx}$ terms. ✓ ½

$$(2x + 3y^2)\frac{dy}{dx} = -(2x + 2y)$$

3 Divide to get the answer. ✓ ½

$$\frac{dy}{dx} = -\frac{2(x+y)}{2x+3y^2}$$

Answer $\frac{dy}{dx} = -\frac{2(x+y)}{2x+3y^2}$

Watch out: Writing $\frac{d}{dx}(2xy) = 2y$ only; the product rule gives $2x\frac{dy}{dx} + 2y$.

14 Short answer — 3 marks

Find the values of a and b such that $f(x) = \begin{cases} 5, & x \le 2 \\ ax + b, & 2 < x < 10 \\ 21, & x \ge 10 \end{cases}$ is continuous.

Formula / idea: Each piece is a polynomial, so only the joining points $x = 2$ and $x = 10$ need checking.

1 At $x = 2$: find $f(2)$, LHL and RHL.

$$f(2) = 5, \quad \lim_{x\to 2^-} f(x) = 5, \quad \lim_{x\to 2^+} (ax + b) = 2a + b$$

2	Continuity at $x = 2$ gives: $2a + b = 5 \quad \cdots(1)$	✓ 1
3	At $x = 10$: find $f(10)$, LHL and RHL. $f(10) = 21, \quad \lim_{x\to 10^-}(ax+b) = 10a + b, \quad \lim_{x\to 10^+} f(x) = 21$	
4	Continuity at $x = 10$ gives: $10a + b = 21 \quad \cdots(2)$	✓ 1
5	Subtract (1) from (2). $8a = 16 \Rightarrow a = 2$	✓ ½
6	Put $a = 2$ in (1). $b = 5 - 4 = 1$	✓ ½

Answer $a = 2, b = 1$

Watch out: Checking only one joining point; f is continuous everywhere only if both $x = 2$ and $x = 10$ work.

Level 3

25 marks

15 **Short answer** 3 marks

If $y = (\tan^{-1} x)^2$, show that $(x^2+1)^2\dfrac{d^2y}{dx^2} + 2x(x^2+1)\dfrac{dy}{dx} = 2$.

Formula / idea: Clear the denominator after the first derivative, then differentiate again.

1	Differentiate w.r.t. x (chain rule). $y_1 = 2\tan^{-1} x \cdot \dfrac{1}{1+x^2}$	✓ 1
2	Cross-multiply. $(1+x^2)\,y_1 = 2\tan^{-1} x$	✓ ½
3	Differentiate again w.r.t. x (product rule). $(1+x^2)\,y_2 + 2x\,y_1 = \dfrac{2}{1+x^2}$	✓ 1
4	Multiply both sides by $(1+x^2)$. Hence proved. $(x^2+1)^2\dfrac{d^2y}{dx^2} + 2x(x^2+1)\dfrac{dy}{dx} = 2$	✓ ½

Answer $(x^2+1)^2\dfrac{d^2y}{dx^2} + 2x(x^2+1)\dfrac{dy}{dx} = 2$ **(proved)**

Watch out: Using the quotient rule on y_1 directly; it works but is long. Cross-multiplying first is safer.

16 **Long answer** 5 marks

If $(x-a)^2+(y-b)^2=c^2$ for some $c>0$, prove that $\dfrac{\left[1+\left(\dfrac{dy}{dx}\right)^2\right]^{3/2}}{\dfrac{d^2y}{dx^2}}$ is a constant independent of a and b.

Formula / idea: Implicit differentiation twice; then use $(x-a)^2+(y-b)^2=c^2$ to simplify.

1 Differentiate the equation w.r.t. x.

$$2(x-a)+2(y-b)\frac{dy}{dx}=0$$

2 Solve for $\dfrac{dy}{dx}$. ✓ 1

$$\frac{dy}{dx}=-\frac{x-a}{y-b} \quad \cdots(1)$$

3 Differentiate (1) again (quotient rule). ✓ ½

$$\frac{d^2y}{dx^2}=-\frac{(y-b)\cdot 1-(x-a)\dfrac{dy}{dx}}{(y-b)^2}$$

4 Put $\dfrac{dy}{dx}$ from (1). ✓ ½

$$\frac{d^2y}{dx^2}=-\frac{(y-b)+\dfrac{(x-a)^2}{y-b}}{(y-b)^2}=-\frac{(y-b)^2+(x-a)^2}{(y-b)^3}$$

5 Use the given equation in the numerator. ✓ ½

$$\frac{d^2y}{dx^2}=-\frac{c^2}{(y-b)^3}$$

6 Now find $1+\left(\dfrac{dy}{dx}\right)^2$ using (1). ✓ 1

$$1+\frac{(x-a)^2}{(y-b)^2}=\frac{(y-b)^2+(x-a)^2}{(y-b)^2}=\frac{c^2}{(y-b)^2}$$

7 Raise to the power $\dfrac{3}{2}$. ✓ ½

$$\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}=\frac{c^3}{(y-b)^3}$$

Taking $y>b$; if $y<b$ the final value is c

8 Divide the two results. ✓ ½

$$\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}=\frac{c^3}{(y-b)^3}\times\frac{(y-b)^3}{-c^2}=-c$$

9	Hence the expression $= -c$, a constant independent of a and b.	✓ ½

Answer **The expression equals $-c$ (or c when $y < b$), a constant independent of a and b.**

Watch out: Not replacing $(x-a)^2 + (y-b)^2$ by c^2; then a and b seem to stay in the answer.

17 **Short answer** 3 marks

If $x^y = e^{x-y}$, prove that $\dfrac{dy}{dx} = \dfrac{\log x}{(1+\log x)^2}$.

Formula / idea: Take $\log$ on both sides; $\log e^k = k$. Make y the subject, then use the quotient rule.

1	Take log on both sides. $y \log x = (x-y)\log e = x - y$	✓ ½
2	Bring the y terms together. $y \log x + y = x \;\Rightarrow\; y(1+\log x) = x$	
3	Make y the subject. $y = \dfrac{x}{1+\log x}$	✓ ½
4	Differentiate w.r.t. x (quotient rule). $\dfrac{dy}{dx} = \dfrac{(1+\log x)\cdot 1 - x \cdot \frac{1}{x}}{(1+\log x)^2}$	✓ 1
5	Simplify the numerator. Hence proved. $\dfrac{dy}{dx} = \dfrac{1+\log x - 1}{(1+\log x)^2} = \dfrac{\log x}{(1+\log x)^2}$	✓ 1

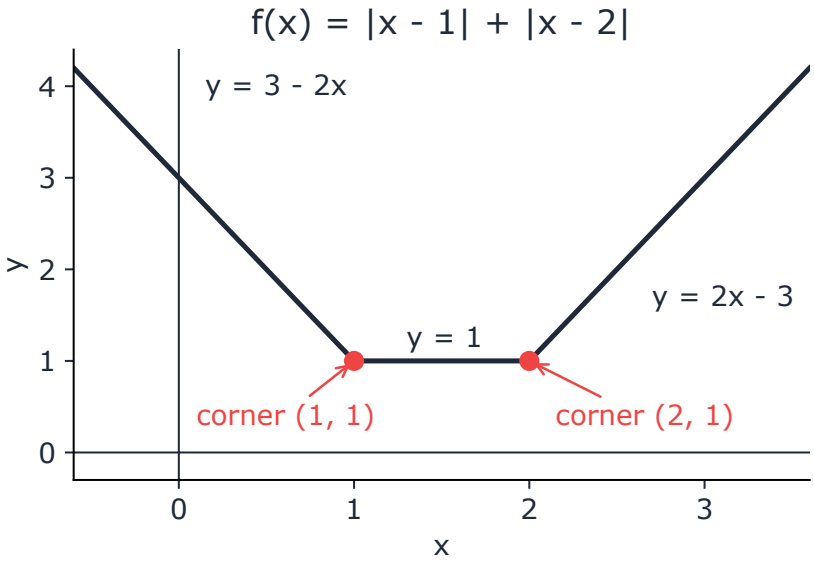
Answer $\dfrac{dy}{dx} = \dfrac{\log x}{(1+\log x)^2}$ **(proved)**

Watch out: Differentiating $y \log x = x - y$ implicitly gives an answer with y in it; making y the subject first avoids this.

18 **Long answer** 5 marks

Show that $f(x) = |x-1| + |x-2|$ is continuous at $x = 1$ and $x = 2$ but not differentiable at these points.

Formula / idea: Remove the modulus piece-wise; then compare LHL, RHL, $f(a)$ for continuity and LHD, RHD for differentiability.



1	Remove the modulus signs on each interval. $f(x) = \begin{cases} 3-2x, & x < 1 \\ 1, & 1 \le x \le 2 \\ 2x-3, & x > 2 \end{cases}$	✓ 1
2	Continuity at $x = 1$: find LHL, RHL and $f(1)$. $\lim\limits_{x\to 1^-}(3-2x) = 1, \quad \lim\limits_{x\to 1^+} 1 = 1, \quad f(1) = 1$	
3	All three are equal, so f is continuous at $x = 1$.	✓ 1
4	Continuity at $x = 2$: find LHL, RHL and $f(2)$. $\lim\limits_{x\to 2^-} 1 = 1, \quad \lim\limits_{x\to 2^+}(2x-3) = 1, \quad f(2) = 1$	
5	All three are equal, so f is continuous at $x = 2$.	✓ 1
6	Differentiability at $x = 1$: find LHD. $\text{LHD} = \lim\limits_{h\to 0} \dfrac{f(1-h)-f(1)}{-h} = \lim\limits_{h\to 0} \dfrac{(1+2h)-1}{-h} = -2$	
7	Find RHD at $x = 1$. $\text{RHD} = \lim\limits_{h\to 0} \dfrac{f(1+h)-f(1)}{h} = \lim\limits_{h\to 0} \dfrac{1-1}{h} = 0$	
8	LHD $\neq$ RHD, so f is not differentiable at $x = 1$. $-2 \neq 0$	✓ 1

9 Differentiability at $x = 2$: find LHD.

$$\text{LHD} = \lim_{h\to 0} \frac{f(2-h) - f(2)}{-h} = \lim_{h\to 0} \frac{1-1}{-h} = 0$$

10 Find RHD at $x = 2$.

$$\text{RHD} = \lim_{h\to 0} \frac{f(2+h) - f(2)}{h} = \lim_{h\to 0} \frac{(1+2h) - 1}{h} = 2$$

$$f(2+h) = 2(2+h) - 3 = 1 + 2h$$

11 LHD $\neq$ RHD, so f is not differentiable at $x = 2$. ✓ 1

$$0 \neq 2$$

12 Hence f is continuous at $x = 1, 2$ but not differentiable at these points.

Answer f **is continuous at** $x = 1$ **and** $x = 2$**; at** $x = 1$**, LHD** $= -2 \neq 0 =$ **RHD; at** $x = 2$**, LHD** $= 0 \neq 2 =$ **RHD, so not differentiable there.**

Watch out: Writing $f(1-h) = 1 - 2h$; in fact $3 - 2(1-h) = 1 + 2h$, so LHD $= -2$.

19 Long answer — 5 marks

If $x = a\left(\cos\theta + \log\tan\frac{\theta}{2}\right)$ and $y = \sin\theta$, then find $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{4}$.

Formula / idea: $\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\cdot\frac{d\theta}{dx}$; never divide $\frac{d^2y}{d\theta^2}$ by $\frac{d^2x}{d\theta^2}$.

1 Differentiate x w.r.t. θ (chain rule on $\log\tan\frac{\theta}{2}$). ✓ ½

$$\frac{dx}{d\theta} = a\left(-\sin\theta + \frac{1}{\tan\frac{\theta}{2}}\cdot\sec^2\frac{\theta}{2}\cdot\frac{1}{2}\right)$$

2 Simplify the second term. ✓ ½

$$\frac{\sec^2\frac{\theta}{2}}{2\tan\frac{\theta}{2}} = \frac{1}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = \frac{1}{\sin\theta}$$

3 Combine the terms. ✓ ½

$$\frac{dx}{d\theta} = a\left(\frac{1-\sin^2\theta}{\sin\theta}\right) = \frac{a\cos^2\theta}{\sin\theta}$$

4 Differentiate y w.r.t. θ. ✓ ½

$$\frac{dy}{d\theta} = \cos\theta$$

5 Divide to get $\frac{dy}{dx}$. ✓ 1

$$\frac{dy}{dx} = \cos\theta\cdot\frac{\sin\theta}{a\cos^2\theta} = \frac{\tan\theta}{a}$$

6 Differentiate again w.r.t. x (chain rule through θ). ✓ 1

$$\frac{d^2y}{dx^2} = \frac{\sec^2\theta}{a}\cdot\frac{d\theta}{dx} = \frac{\sec^2\theta}{a}\cdot\frac{\sin\theta}{a\cos^2\theta}$$

7 Simplify.
$\frac{d^2y}{dx^2} = \frac{\sin\theta}{a^2\cos^4\theta}$ ✓ ½

8 Put $\theta = \frac{\pi}{4}$: $\sin\theta = \frac{1}{\sqrt{2}}$, $\cos^4\theta = \frac{1}{4}$.
$\left.\frac{d^2y}{dx^2}\right|_{\theta=\frac{\pi}{4}} = \frac{1/\sqrt{2}}{a^2\cdot\frac{1}{4}} = \frac{4}{\sqrt{2}\,a^2} = \frac{2\sqrt{2}}{a^2}$ ✓ ½

Answer $\frac{d^2y}{dx^2} = \frac{2\sqrt{2}}{a^2}$ **at** $\theta = \frac{\pi}{4}$

Watch out: Forgetting to multiply by $\frac{d\theta}{dx}$ when differentiating $\frac{\tan\theta}{a}$ w.r.t. x.

20 Case study — 4 marks

Case Study: While designing a smooth ramp, an engineer models its profile by $f(x) = \begin{cases} x^2+3x+p, & x \le 1 \\ qx+2, & x > 1 \end{cases}$. For a smooth ramp, f must be differentiable at $x = 1$.

(i) Write the condition on p and q for f to be continuous at $x = 1$. — 1 mark

(ii) Find the left-hand derivative and right-hand derivative of f at $x = 1$. — 1 mark

(iii) Find the values of p and q. — 2 marks

Formula / idea: Differentiable at $x = 1$ needs continuity there and LHD = RHD.

(i) — 1 mark

1 Find $f(1)$ (equals LHL) and the RHL.
$f(1) = 1+3+p = 4+p, \qquad \lim_{x\to1^+}(qx+2) = q+2$ ✓ ½

2 For continuity at $x = 1$, equate them.
$4+p = q+2$ i.e. $q = p+2$ ✓ ½

Answer (i) $4+p = q+2$, **i.e.** $q = p+2$

(ii) — 1 mark

1 LHD at $x = 1$ from the left piece.
$\text{LHD} = \lim_{h\to0}\frac{(1-h)^2+3(1-h)+p-(4+p)}{-h} = \lim_{h\to0}\frac{h^2-5h}{-h} = 5$ ✓ ½

2 RHD at $x = 1$ from the right piece.
$\text{RHD} = \lim_{h\to0}\frac{q(1+h)+2-(4+p)}{h} = \lim_{h\to0}\frac{qh}{h} = q$
Using (i): $q+2-(4+p) = 0$ ✓ ½

Answer (ii) **LHD** $= 5$, **RHD** $= q$

(iii) 2 marks

1	For differentiability, LHD $=$ RHD. $q = 5$	✓ 1
2	Put $q = 5$ in the condition from (i). $4 + p = 5 + 2 \Rightarrow p = 3$	✓ 1

Answer (iii) $p = 3, q = 5$

Watch out: Equating only LHD and RHD; continuity at $x = 1$ is also needed to find p.